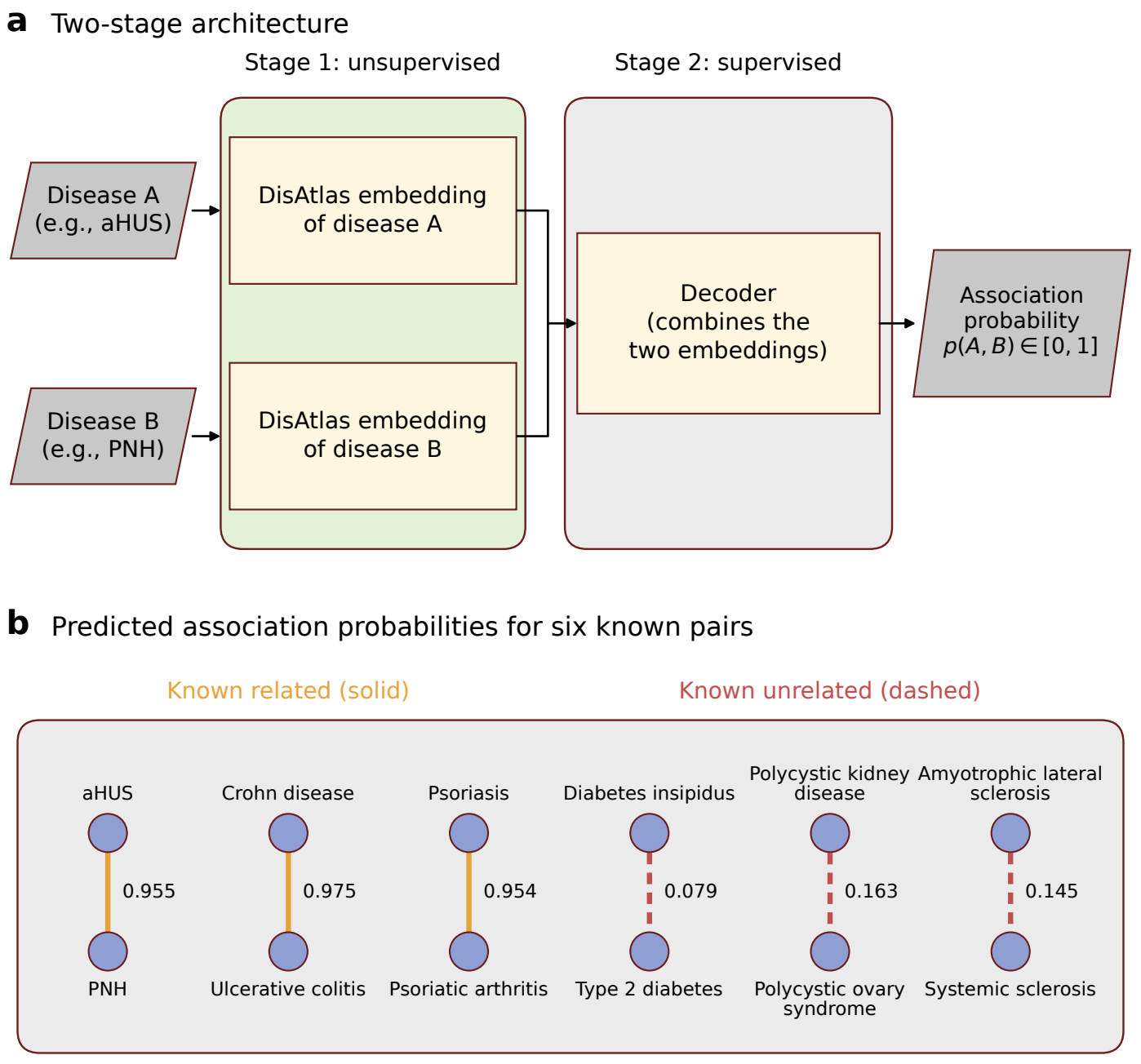
a
Two-stage architecture
Stage 1: unsupervised
Stage 2: supervised
Disease A
(e.g., aHUS)
Disease B
(e.g., PNH)
DisAtlas embedding
of disease A
DisAtlas embedding
of disease B
Decoder
(combines the
two embeddings)
Association
probability
$p(A, B) \in [0, 1]$
b
Predicted association probabilities for six known pairs
Known related (solid)
Known unrelated (dashed)
aHUS
0.955
PNH
Crohn disease
0.975
Ulcerative colitis
Psoriasis
0.954
Psoriatic arthritis
Diabetes insipidus
0.079
Type 2 diabetes
Polycystic kidney disease
0.163
Polycystic ovary syndrome
Amyotrophic lateral sclerosis
0.145
Systemic sclerosis